INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

STRAINERS, SELF-CLEANING, BACKFLUSH TYPE (MANUAL OR AUTOMATED)

The General Service Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers both manual and automated backflush type self–cleaning strainers.

1.1 Intended use. The strainers specified are intended for installation in seawater piping systems on the discharge side of the pump.

2. CLASSIFICATION. Backflush type self-cleaning strainers shall be of the following types, classes, styles, and sizes as specified (see 7.1).

2.1 Type.

a. Type I - Manual.

b. Type II - Automated.

2.2 Class.

a. Class 150 - 150 pounds per square inch gage (psig) pressure rating.

b. Class 250 - 250 psig pressure rating.

2.3 Style.

a. Style 1 - Rotational divider, Sizes 2, 3, 4, 6, 8, 10, 12, 14, 16, 20, and 24 inches.

b. Style 2 - Rotational backwash arm, Sizes 2, 3, 4, 6, 8, 10, 12, 14, 16, 18, 20, and 24 inches.

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3. SALIENT CHARACTERISTICS.

3.1 Codes and standards. Strainers shall comply with the applicable requirements of table I.

Table I. Reference data.

Equipment/Characteristic	Description
Fuel type	Seawater
Operating pressure	150, 250 psi
Temperature range	70 to 150 °F
Piping connections: Inlet, outlet, and screen compartment Vent, drain, and gage	 MIL-PRF-20042 raised face flanges SAE J1926/1
Screen	Table VII
Pressure resistance	FCI 78-1
Flow capacity	Table III in accordance with ISA-75.02
Shock resistance	MIL-S-901, Grade A
Type II strainer only: Motor Control panel	 MIL-M-17059 MIL-DTL-2212

3.2 Design and construction. Each strainer shall be of the type, class, style, and size as specified (see 2.1 through 2.3). The self-cleaning action shall remove particulate contaminants from the seawater system without interrupting the operation of that system. The strainer screen shall be cleaned in segments such that the particulate contaminants are flushed free of the strainer.

3.3 Material. The material requirements for all strainer components shall be in accordance with table II.

Table II. Materials required for strainers.

Part	Material	Applicable Documents	Remarks
Bodies, covers	Copper-nickel	ASTM B369	C96400
	NiAl bronze	ASTM B148	C95800
Collar, handle	Cast tin bronze	ASTM B61	C92200
	CRES	ASTM A276/A276M	C31600
Self-cleaning mechanism	NiCu alloy	ASTM B127	N04400
	NiAl bronze	ASTM B148	C95800
Threaded fasteners	NiCu alloy	ASTM F467	N04400, N04405
		ASTM F468	
O-ring	Fluorocarbon	SAE-AS3209	For all except threaded bosses
		SAE-AS3581	Threaded bosses
Strainer screen	CuNi plate	ASTM B122/B122M	C71500
	NiCu plate	ASTM B127	N04400, N04405

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3.4 Strainer screen. The strainer screen shall be made of one unit. The strainer screen shall be of the perforated plate or plate type construction as specified (see 7.1). All seam welds shall be continuous.

3.4.1 Strainer screen compression strength test. The strainer screen shall be tested by applying a differential pressure (outside-inside) across the strainer screen equal to 100 percent of the specified strainer design pressure for the applicable type and class, using water for a duration of 5 minutes. Differential pressure shall be applied in a rapid manner, for 1 minute or less. Strainer screens that distort, stretch, or exhibit other weaknesses shall be rejected.

3.4.2 Strainer screen expansion strength test. The strainer screen shall be tested by applying a differential pressure (inside-outside) across the strainer screen equal to 100 percent of the specified strainer design pressure for the applicable type and class, using water for a duration of 20 minutes. Differential pressure shall be applied in a rapid manner, for 1 minute or less. Strainer screens that distort, stretch, or exhibit other weaknesses shall be rejected.

3.5 Actuation. Each strainer shall be fitted with a shaft along its rotational axis, which penetrates the compartment cover through a watertight seal. Outside the housing, the shaft shall be capable of being fitted with either a handle (Type I) or an electric motor actuator (Type II). Each Type I strainer shall be sized and configured so that maximum required operating force is less than 50 pounds. For certain hazard areas, such as installation in a JP-5 pump room, Type II strainer motor actuator shall be explosion-proof.

3.6 Self-cleaning mechanism.

3.6.1 Style 1. The Style 1 strainer screen is divided by a vertical partition which yields two halves of a cylindrical strainer screen. At the bottom of each half, there is a drain connection which will be used to connect backflush valves at installation. Backflushing is accomplished by rotating the divider handle 180 degrees, diverting flow to the clean half of the strainer screen, and then opening the drain valve of the dirty half. A portion of the fluid flows back through the dirty strainer screen freeing the contaminant and allowing the debris to be flushed from the strainer. Fluid used to flush the debris out of the strainer shall not exceed 15 percent of the total flow through the strainer assembly. Strainers shall be fitted with two flushing connections; one for each partitioned straining compartment.

3.6.2 Style 2. The Style 2 strainer screen is cleaned by a hollow backwash arm which rotates inside the strainer screen. Backflushing is accomplished by opening the drain valve and diverting fluid flow back through a portion of the strainer screen. The debris flows into the hollow backwash arm and out of the strainer. The backwash arm then rotates to the next portion of the strainer screen. Fluid used to flush the debris out of the strainer shall not exceed 15 percent of the total flow through the strainer assembly.

3.7 Covers. Covers shall be flanged and attached to the body by bolts. Each cover shall effect a positive seal on an O-ring.

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3.8 Functional performance. Unless otherwise specified (see 7.1), design flow capacity and clean pressure drop shall be as specified in table III and design temperature shall be 150 °F.

Table III. Functional performance.

Size	Clean Pressure Drop at Flow Rate	
	(psi)	(gpm)
2	2	80
3	2	175
4	2	325
6	2	750
8	2	1400
10	2	2000
12	2	3000
14	2	4000
16	2	5000
18	2	7000
20	2	9000
24	2	13000

3.8.1 Operational cycling tests. The strainer shall be tested for a total of 600 hours at design pressure and flow rates. Strainers shall be backflushing once every hour during testing period. At the completion of the 600 hours, compliance with the required operating force of 50 pounds or less and zero leakage shall be demonstrated. Upon satisfactory completion of this test, the unit shall be refurbished to new condition before being offered for delivery.

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3.9 Envelope dimensions. Envelope dimensions (inches) shall be in accordance with table IV.

Table IV. Envelope dimensions.

Size	Style 1				Style 2			
	Flange-to-Flange	Width	Height	Screen Removal	Flange-to-Flange	Width	Height	Screen Removal
2	14.00	11.00	20.25	24.50	18.25	12.00	28.00	37.00
3	15.00	12.00	20.25	24.50	18.25	12.00	28.00	37.00
4	19.00	14.00	23.50	28.00	18.25	12.00	28.00	37.00
6	24.00	18.00	28.50	34.25	29.00	21.25	39.00	50.00
8	29.50	22.75	25.50	42.50	27.00	21.25	39.00	50.00
10	35.50	29.00	35.25	53.25	39.50	31.00	54.00	74.00
12	42.50	33.50	51.25	61.25	37.75	31.00	54.00	74.00
14	47.50	37.00	58.25	73.50	45.50	37.00	55.50	82.00
16	54.25	40.25	64.25	82.75	45.50	37.00	55.50	82.00
18	----	----	----	----	55.00	44.25	71.50	85.00
20	55.50	51.75	76.25	99.25	52.00	44.25	71.50	85.00
24	76.00	60.00	90.00	107.00	65.00	65.00	90.00	113.00
NOTE: All dimensions are in inches.								

3.10 Workmanship. The strainer body and covers shall be of uniform quality and condition, free from blow holes, porosity, hard spots, shrinkage defects, cracks, and other defects. All surfaces shall be smooth and well cleaned. The inside surfaces of strainers shall be well cleaned and free from sharp edges.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Prohibited material. Zinc or cadmium plating shall not be used on any part of the strainer.

4.3 Mercury exclusion. The strainer shall be free of mercury contamination. During the manufacturing process, tests, and examination, the product to be offered for acceptance shall not come in direct contact with mercury or any of its components, nor with any mercury containing device employing a single boundary of containment.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same production offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Additional inspection. Unless otherwise specified in the contract or order (see 7.1), where other specifications form a part of this CID, sampling, examination, and tests shall be conducted as required by the pertinent specification.

6. PACKAGING.

6.1 General requirement. Commercial packaging shall be in accordance with the requirements of ASTM D3951.

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6.2 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage, and storage.

7. NOTES.

7.1 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN (see 7.6).

b. Type, class, style, and size required (see 2.1, 2.2, and 2.3).

c. Strainer screen design and characteristics (see 3.4).

d. The required design flow capacity, design pressure, design temperature, and clean pressure drop if not as specified (see 3.8 and table III).

e. Product conformance provisions (see 5.2).

f. Whether repair parts are required (see 7.4.1).

7.2 Source of documents.

7.2.1 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.2.2 Defense specifications and standards. Defense specifications and standards are available online at http://quicksearch.dla.mil.

7.2.3 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

7.2.4 Fluid Control Institute (FCI). FCI standards are available from the Fluid Controls Institute, Inc., 1300 Sumner Avenue, Cleveland, OH 44115 or online at http://www.fluidcontrolsinstitute.org/.

7.2.5 International Society of Automation (ISA). ISA standards are available from the International Society of Automation, 67 T.W. Alexander Dr., P.O. Box 12277, Research Triangle Park, NC 27709 or online at www.isa.org.

7.2.6 SAE International. SAE standards are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.

7.3 Sub-contracted material and parts. The packaging or delivery preparation requirements of referenced documents listed in 7.2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

7.4 Ordering spare or repair parts. When ordering spare parts or repair parts for the equipment covered by this CID, the contract should state that such spare parts and repair parts should meet the same requirements as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

7.4.1 Repair parts and repair parts boxes. Complete sets of repair strainer screens for each type and class should be furnished as follows:

a. One set for each of the ten initial strainers required.

b. One set for each two additional strainers required.

c. Cover O-rings - four for every strainer supplied.

d. Other O-rings and seals - one set for every ten strainers supplied.

When specified (see 7.1), 7.4.1.c and 7.4.1.d should be furnished in repair boxes.

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7.5 Preinstallation instruction. A set of instructions covering the preinstallation of the equipment should be furnished. Instructions should include all information necessary to return the unit to active status, such as, but not limited to, the addition of lubricants prior to operation, flushing of lines, removal of greaseproof barrier, and the location of detached components. Instructions should be packaged in a transparent waterproof plastic bag. Closure should be by heat sealing. The shipping container in which the instructions are packed should be so marked.

7.6 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN is a definitive number which corresponds to the type, class, style, size, and screen perforation of strainers covered by this CID. The CID number, the type and class code letter, the style and size code number, and screen perforation code letter are combined to form a definitive PIN. PINs for the strainers are assigned as follows:

AA59008	**X**	**XX**	**X**
CID Number	Type and Class (see 7.6.1)	Style and Size (see 7.6.2)	Screen Perforation (see 7.6.3)

7.6.1 Type and class. The type and class of strainer (see 2.1 and 2.2) are identified by a single letter (see table V).

Table V. Code letter for type and class.

Class	**Type**	
	Type I	**Type II**
Class 150	A	B
Class 250	C	D

7.6.2 Style and size. The style and size of strainer (see 2.3) are identified by a double-digit number (see table VI).

Table VI. Code number for size and style.

Size	**Style**	
	Style 1	**Style 2**
2	01	02
3	03	04
4	05	06
6	07	08
8	09	10
10	11	12
12	13	14
14	15	16
16	17	18
18	--	20
20	21	22
24	23	24

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7.6.3 Screen perforation. The screen perforation of strainers is identified by a single letter (see table VII).

Table VII. Code letter for screen perforation.

Size (Inch)	System Particle Size Tolerance Range (Inch)	Code
¼	0.47 to 0.62	A
3/16	0.31 to 0.47	B
⅛	0.24 to 0.31	C
3/32	0.15 to 0.24	D
1/16	0.09 to 0.15	E
1/32	0.06 to 0.09	F

7.6.4 Example PIN. AA59008C01B is 2-inch rotational divider (Style 1) strainer with a 250 psig pressure rating (Class 250) and a screen perforation of 3/16 inch fitted with a manual hand wheel actuator (Type I).

7.7 Key words.

Cleaning
Filtration
Motor
Screen

MILITARY INTERESTS:

Custodians:
Army – MR
Navy – SH
Air Force – 99

Preparing Activity:
Navy – SH
(Project 4730-2017-032)

Review Activities:
Navy – SA
Air Force – 71
DLA – CC

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